



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

EEG/MEG 2: Spatial Resolution

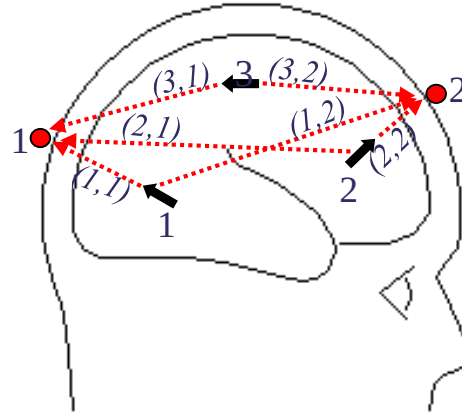
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Spatial Resolution of EEG/MEG – Basic Concepts

Resolution Matrix, Point-Spread and Cross-Talk Functions (PSFs and CTFs)

The EEG/MEG Forward Problem



$$\begin{array}{c}
 \text{data} \quad \text{"leadfield"} \quad \text{dipoles} \\
 \begin{array}{c} 1 \\ \bullet \end{array} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0.3 \\ 0 & 1 & -0.3 \end{pmatrix} \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} \begin{array}{c} \nwarrow 1 \\ \nearrow 2 \\ \nwarrow 3 \end{array} \\
 \bullet 2
 \end{array}
 \xrightarrow[\text{inversion}]{?}
 \begin{array}{c}
 \text{dipoles} \quad \text{inverse} \quad \text{data} \\
 \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} = \begin{pmatrix} 1.5034 & 0.1241 \\ 0.2483 & 0.9379 \\ 0.8276 & -0.2069 \end{pmatrix} * \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \begin{array}{c} \bullet 1 \\ \bullet 2 \end{array}
 \end{array}$$

$$j_1 + j_2 = 1$$

under-determined problem, no unique solution

$$\mathbf{d} = \mathbf{L}\mathbf{j}$$

$\mathbf{d}$: data ($n_{\text{sensors}} \times 1$) $\mathbf{L}$: "leadfield" ($n_{\text{sensors}} \times n_{\text{dipoles}}$), $\mathbf{j}$: dipoles ($n_{\text{dipoles}} \times 1$)

Usually $n_{\text{dipoles}} \gg n_{\text{sensors}}$.

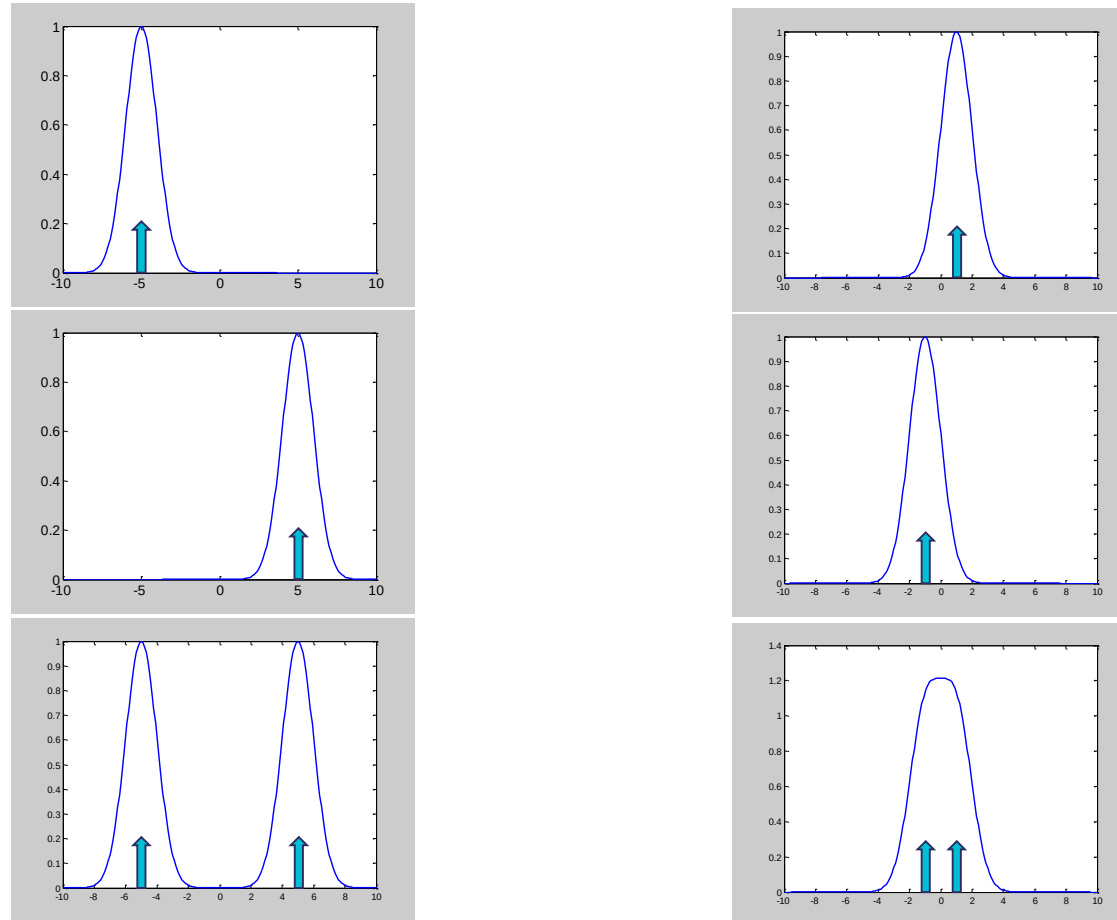
Let's Start Again: The “Blurry Image” Analogy

Just because the brain is complicated doesn't mean source estimation has to be complicated



Linear Methods Can Easily Tell Us If They Do What We Want

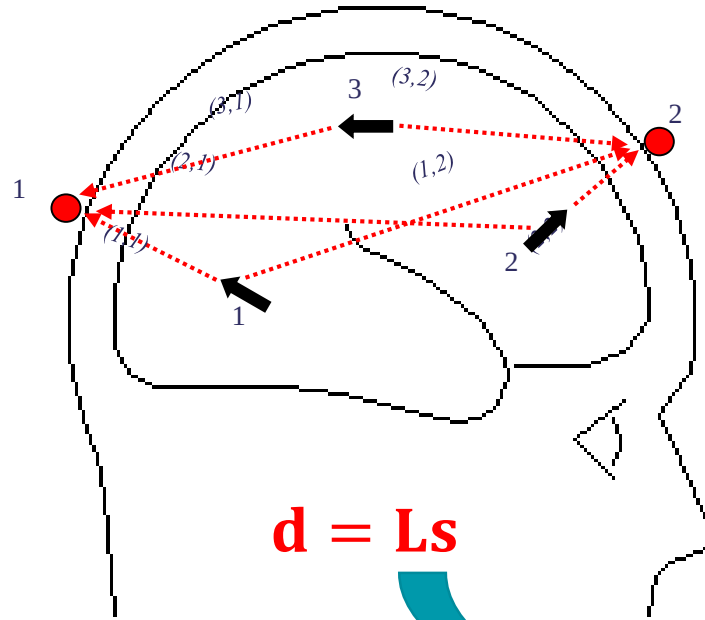
Superposition Principle



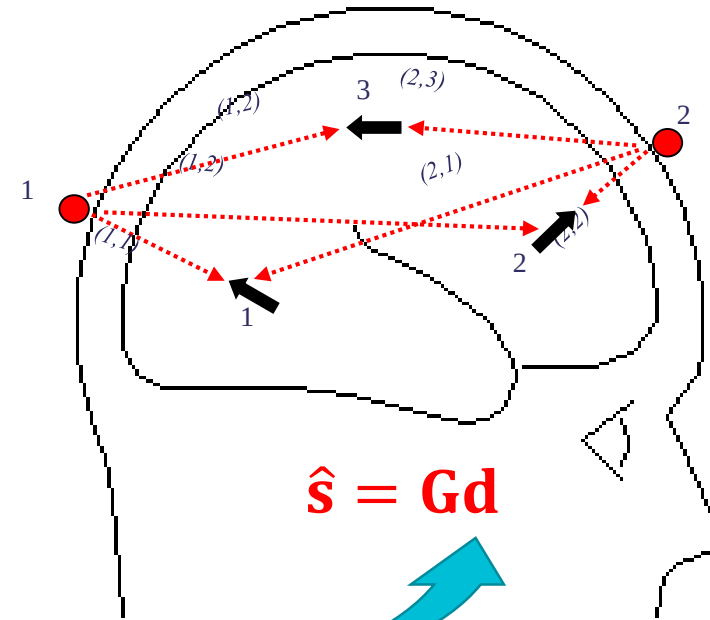
If you know the behaviour for point sources,
you can predict the behaviour for complex sources.

The Resolution Matrix

Forward Problem



Linear Inverse Problem



$$\hat{s} = GLs \stackrel{\text{def}}{=} Rs$$

Relationship between estimated and true source distribution.

Creating an Optimal Resolution Matrix

$$\hat{\mathbf{s}} = \mathbf{R}\mathbf{s}$$

The closer $\mathbf{R}$ is to the identity matrix, the closer our estimate is to the true source.

Therefore, let us minimise the difference between $\mathbf{R}$ and the identity matrix in the least-squares sense:

$$\|\mathbf{R} - \mathbf{I}\|_2 = \min$$

This leads to the **Minimum Norm Estimator (MNE)**:

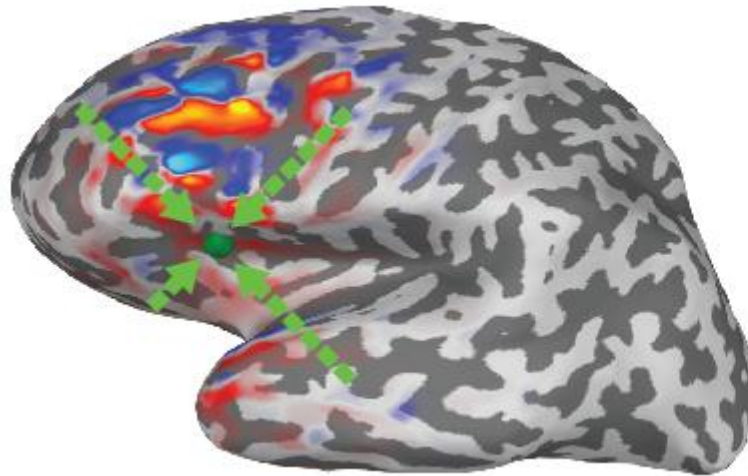
$$\mathbf{G}_{MN} = \mathbf{L}^T (\mathbf{L}\mathbf{L}^T)^{-1}$$

Its resolution matrix $\mathbf{R}_{MN} = \mathbf{L}^T (\mathbf{L}\mathbf{L}^T)^{-1} \mathbf{L}$ is symmetric.

Spatial Resolution / Leakage:

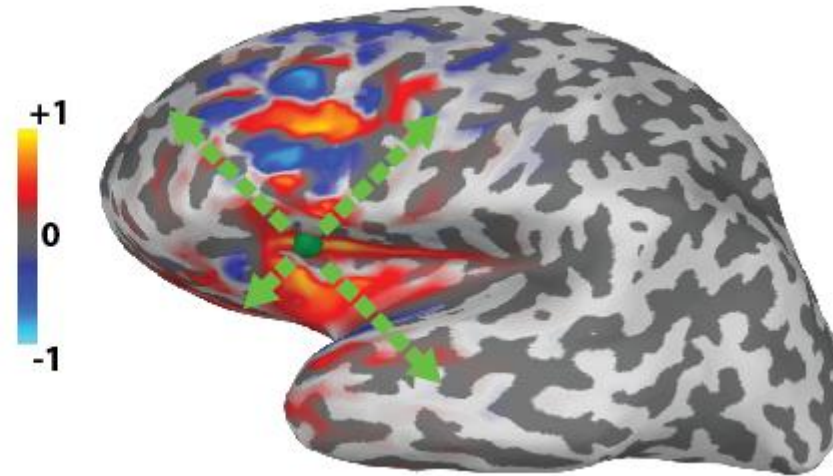
Point-Spread and Cross-Talk

Cross-Talk Function
(CTF)



How other sources may affect the estimate for this source

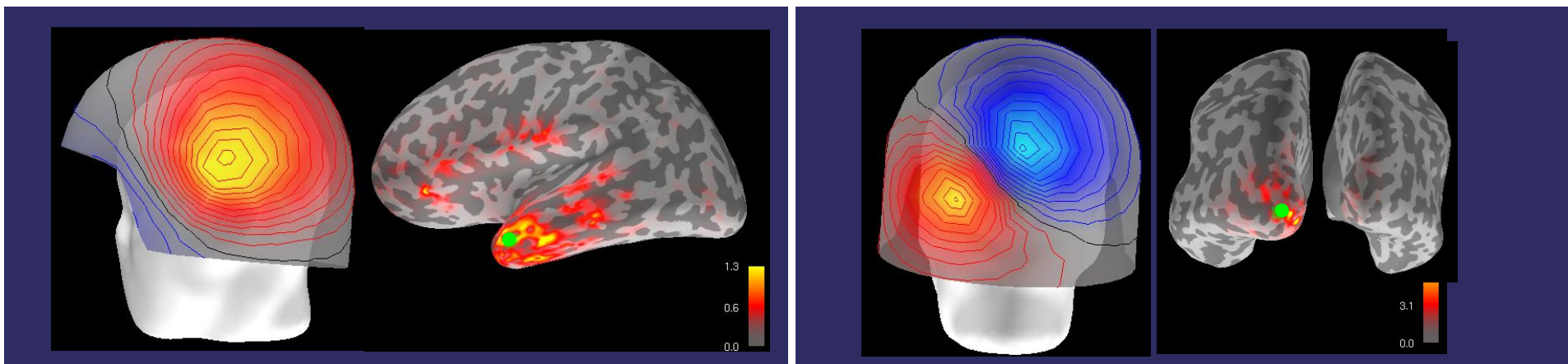
Point-Spread Function
(PSF)



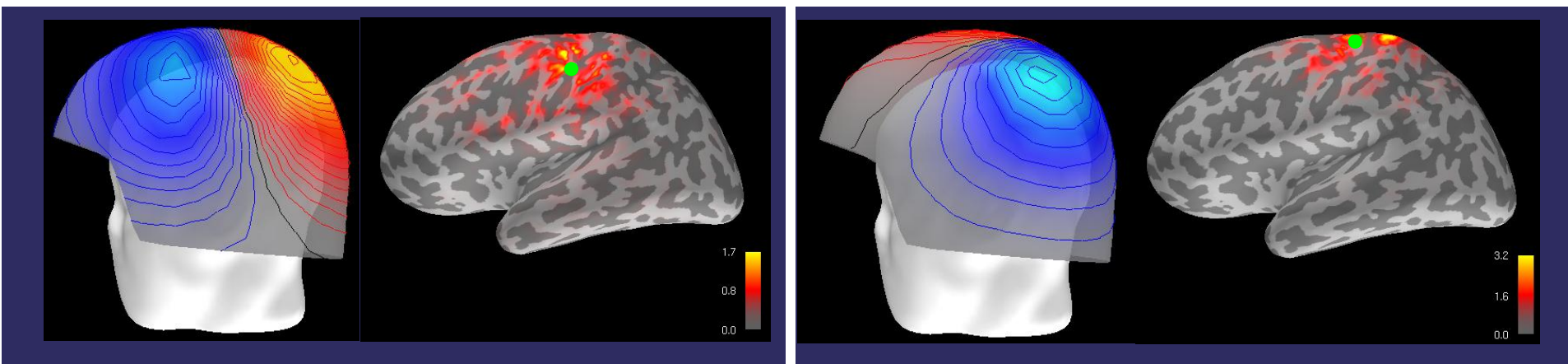
How this source affects estimates for other sources

PSFs and CTFs for Some ROIs

For MNE, PSFs and CTFs turn out to be the same

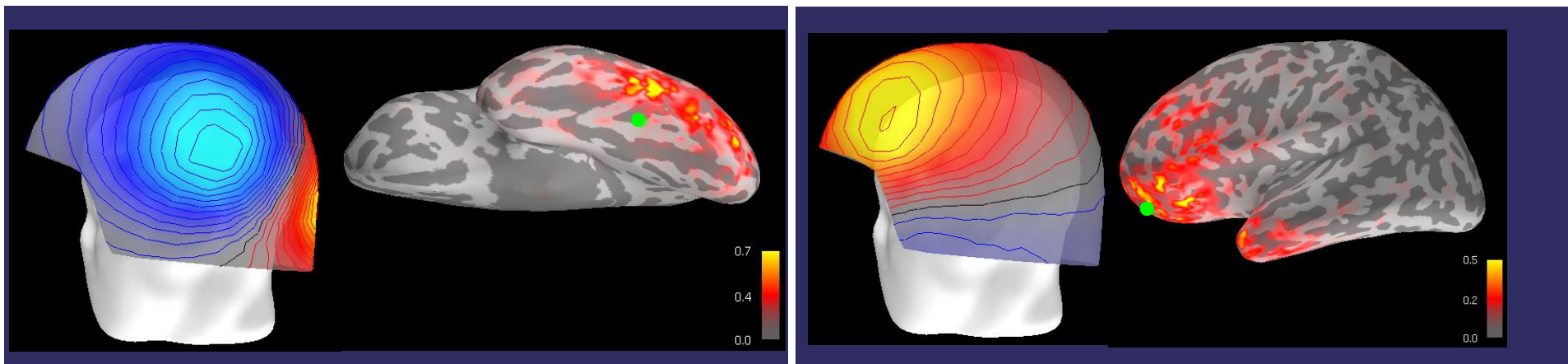


Good

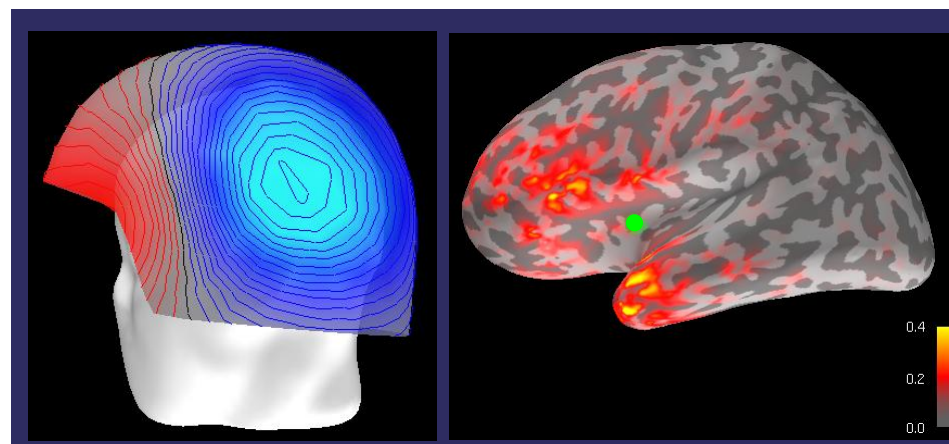


PSFs and CTFs for Some ROIs

For MNE, PSFs and CTFs turn out to be the same

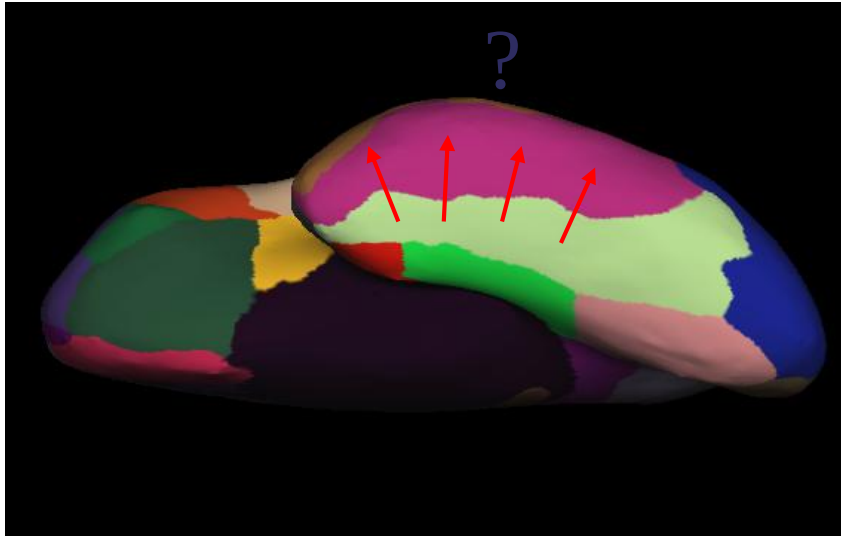


Less good

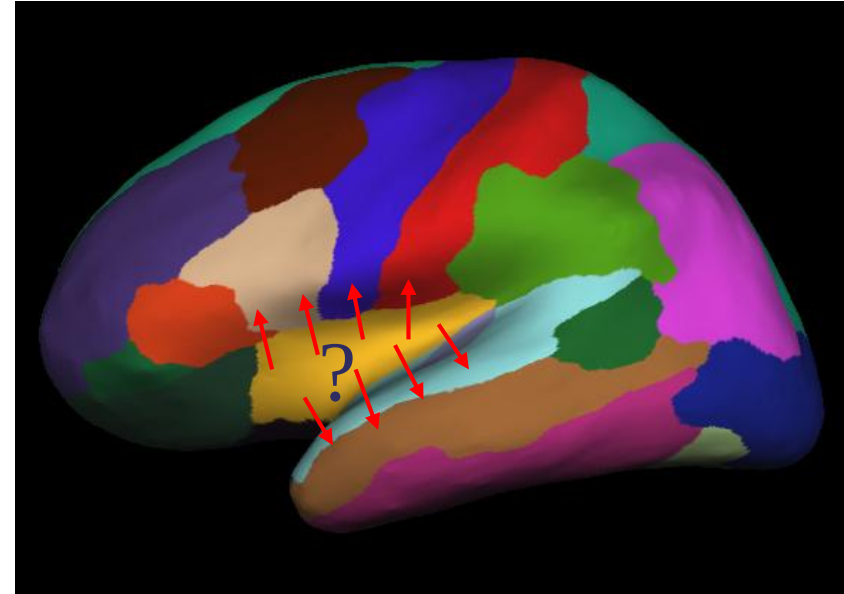


Localisation Bias Has Consequences for ROI analysis

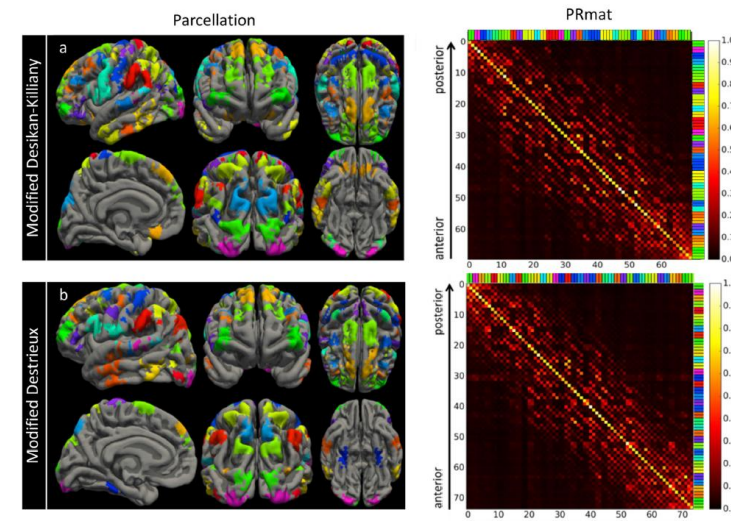
PSFs/CTFs Can Tell You How It Looks Like



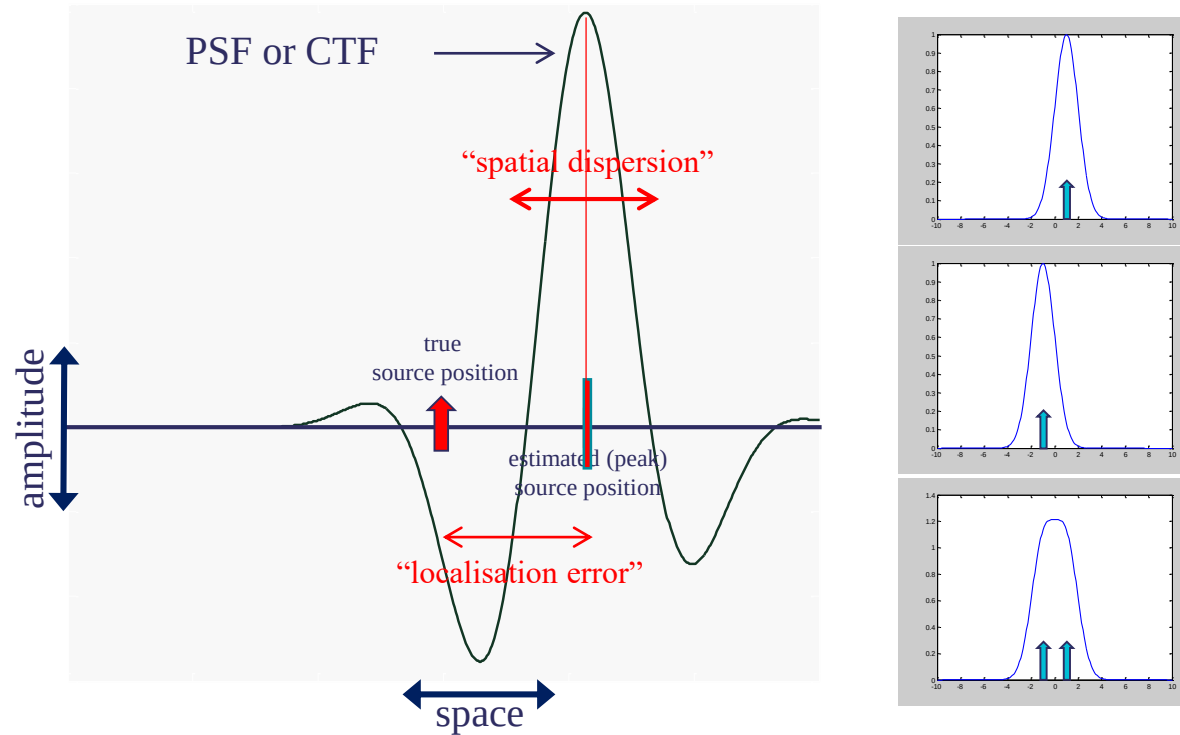
Desikan-Killiany Atlas parcellation



Adaptive cortical parcellation based on resolution matrix are possible: Farahibozorg/Henson/Hauk NI 2018
<https://pubmed.ncbi.nlm.nih.gov/28893608/>

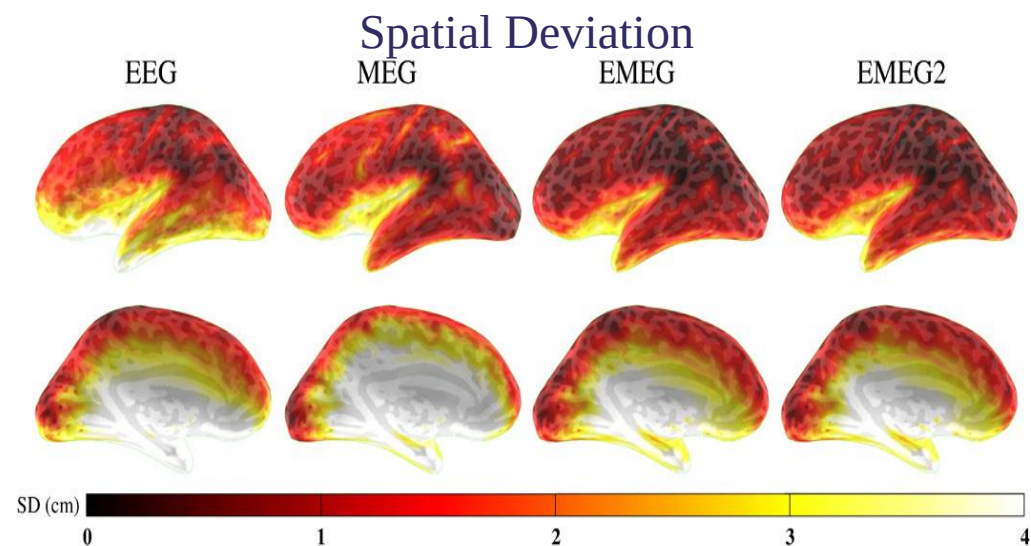
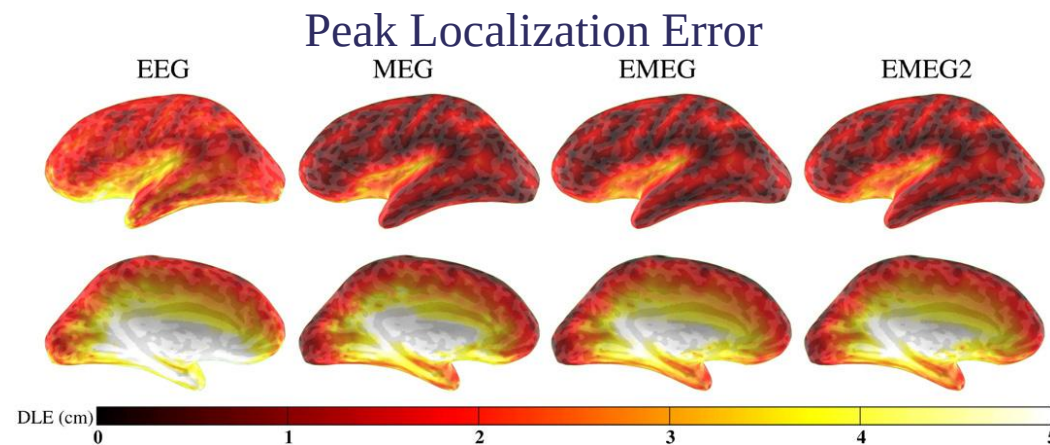


Quantifying Resolution From PSFs and CTFs



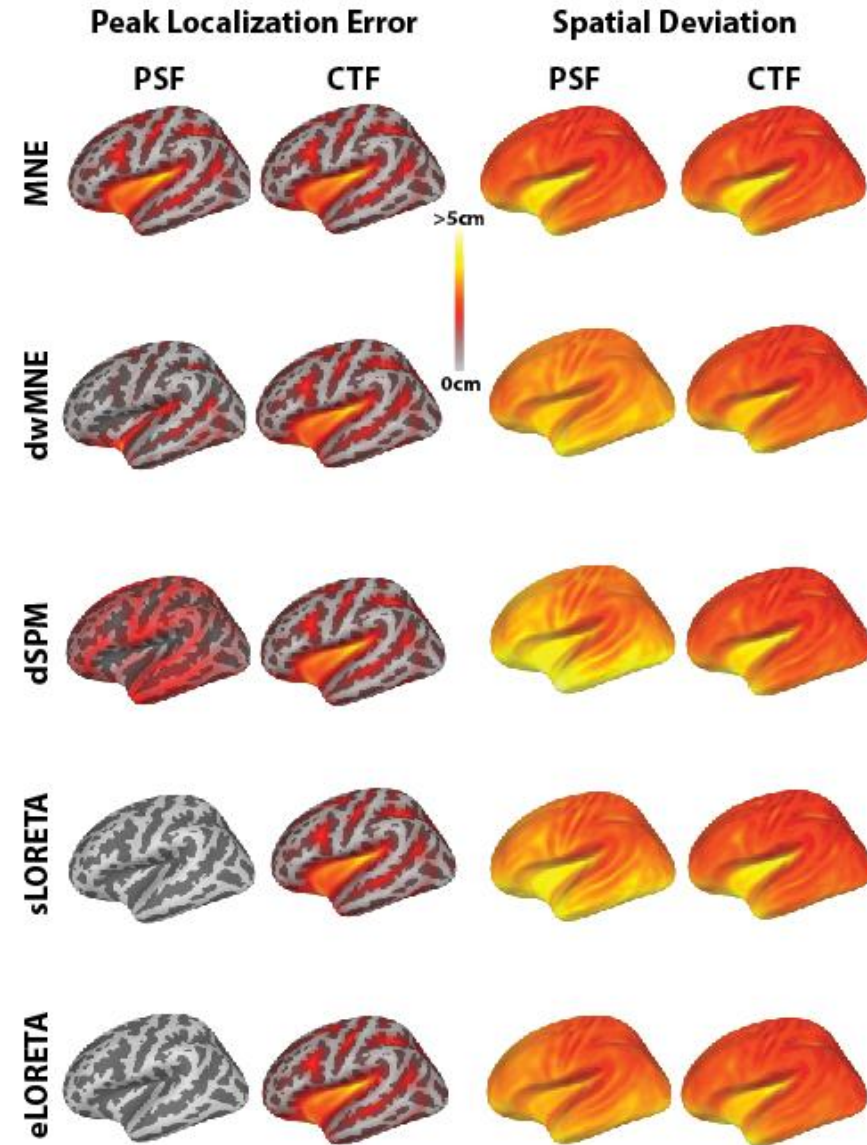
It's not just peak localisation that counts,
but also spatial extent of the distribution.

Whole-Brain Maps of Resolution Metrics

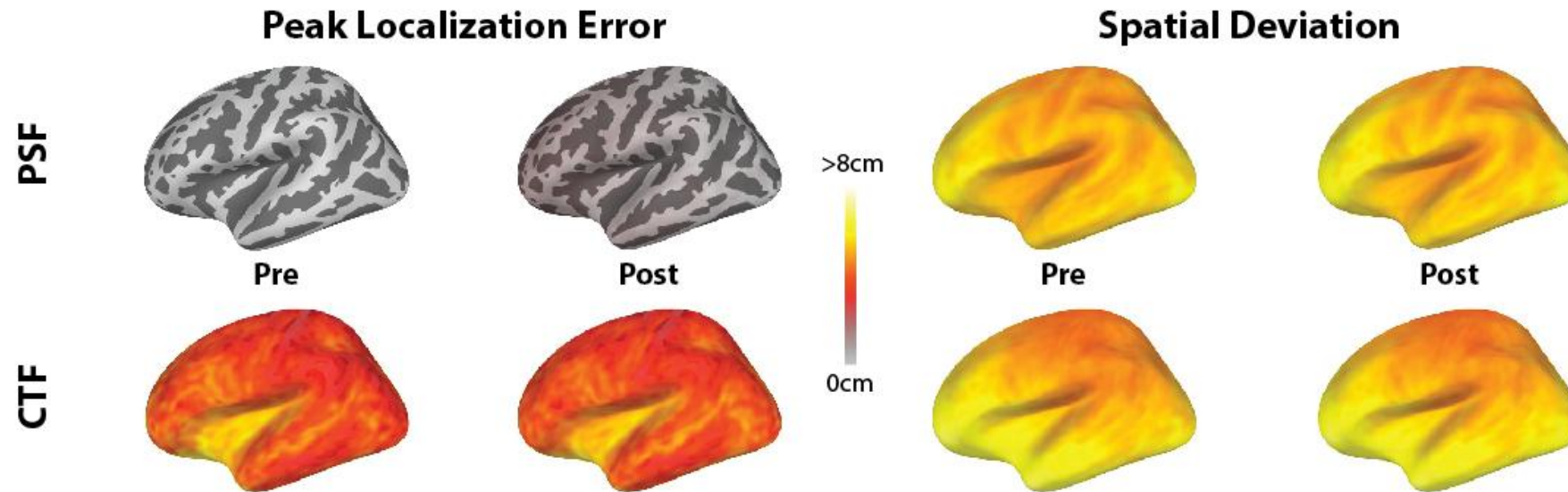


Combining EEG and MEG improves spatial resolution.

Comparing Estimators – MNE-type methods



Comparing Estimators – Beamformers



Thank you